

Student 5 **DANIMIZA V** 192421363RC

4 / 4 submitted

Q1. String Character Frequency Counter

CODE

```
s = input().lower()
count = {}
for ch in s:
    if ch != " ":
        if ch in count:
            count[ch] += 1
        else:
            count[ch] = 1
for ch in sorted(count):
    print(ch,"->",count[ch])
```

INPUT

Hello World

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()
count = {}
for ch in s:
    if ch != " ":
        if ch in count:
            count[ch] += 1
        else:
            count[ch] = 1
for ch in sorted(count):
    print(ch,"->",count[ch])
```

INPUT

Hello World

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q3. String Compression and Run-Length Encoding**CODE**

```
s = input()
c = ""
i = 0
while i < len(s):
    count = 1
    while i < len(s)-1 and s[i] == s[i+1]:
        count += 1
        i += 1
    c += s[i] + (str(count) if count > 1 else "")
    i += 1
if len(c) < len(s):
    print("compressed:",c)
else:
    print("original:",s)
```

INPUT

aabbccdddee

OUTPUT

compressed: a2b3c2d3e2

Q4. Advanced Text Statistics Analyser**CODE**

```
from collections import counter
import re
t = input()
w = re.findall(r'\b\w+\b',t.lower())
s = re.split(r'[.!?]+' ,t.strip())
stop = {"the","is","a","an","in","of","and","to","for"}
f = [i for i in w if i not in stop]
c = counter(f)
print("total words:",len(w))
print("sentences:",len([i for i in s if i]))
print("paragraphs:",1)
print("avg word length:",round(sum(map(len,w))/len(w),2))
print("avg sentence length:",round(len(w)/max(1,len([i for i in s if i])),2))
print("top 5 words:",c.most_common(5))
```

INPUT

python is a powerful programming language

OUTPUT

```
File "C:\Windows\TEMP\sandbox_192421363RC_6a625da88fabf3.47163693\main.py", line 1
    form collections import counter
    ^^^^^^^^^^^^^
SyntaxError: invalid syntax
```